

Chapter 5:

Observables, Operators, Eigenvectors, and Eigenvalues

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Observables and Operators

- Observables are something we can observe and measure.
- We say that if we measure a state, it will collapse to one of the basis states corresponding to an observable. Then it means that the basis state (vector) must be a value we can understand classically.
- To measure the spin in the $\hat{z}$ direction, we need to apply a magnetic field in the $\hat{z}$ direction, and through some processes (measurement, photon emission), we measure its energy.
- If it has lower energy, we know it collapses to the $|\uparrow\rangle$ state. If it has higher energy, we know it collapses to the $|\downarrow\rangle$ state.
- For measurement the spin in the $\hat{x}$ direction, if the energy level measured is low, we know it collapses to the $|\leftarrow\rangle$ state. If the energy level measured is high, we know it collapses to the $|\rightarrow\rangle$ state.
- We see that every measurement corresponds to an observable and each measurement result must be a basis state corresponding to that observable/measurement.
- For the position observable, the basis contains the basis states of (x, y, z) , and since x, y, z can take any values, it has an infinite number of basis states.
- In summary, a basis formed by the basis states corresponding to an observable is a convenient choice in the formulation of the quantum computing problems.

Observables and Operators

- There are two formalisms in quantum mechanics. One is the wave quantum mechanics by Erwin Schrödinger or **Schrödinger Equation**, which describes how a quantum state evolves.
- The other formalism is the **matrix quantum mechanics**. It was proposed independently by Werner Heisenberg. It has exactly the same content as the wave quantum mechanics including Schrödinger equation, but all in matrix form.
- The matrix quantum mechanics is very useful in quantum computing. It is particularly powerful when we combine it with Dirac's bra-ket notation.
- We represent quantum mechanical states as vectors. And we use an operator to describe the evolution of a quantum state.
- An operator operates on a vector. It transforms the vector from one to another.
- The transformation can be stretching and rotation, etc.
- In quantum computing, it is nothing but just the rotation of vectors (states) in the hyperspace.
- In the matrix quantum mechanics, *operators are represented as matrix*.

Eigenvalues and Eigenvectors

- There is a special type of operators that correspond to some observables/measurements. For example, the operator corresponding to the measurement of electron spin in the $\hat{z}$ direction is given as:

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- If we try to operate on the two possible measurable outputs, i.e. $|\uparrow\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\downarrow\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$$\sigma_z |\uparrow\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times 0 \\ 0 \times 1 + -1 \times 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\uparrow\rangle$$

$$\sigma_z |\downarrow\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \times 0 + 0 \times 1 \\ 0 \times 0 + -1 \times 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix} = -|\downarrow\rangle$$

- Both $|\uparrow\rangle$ and $|\downarrow\rangle$ are unchanged when acted by σ_z , with $|\uparrow\rangle$ and $|\downarrow\rangle$ scaled by 1 and -1, respectively. (Note that while it scales $|\downarrow\rangle$ by -1, the vector is still the same in quantum mechanics.)
- So, $|\uparrow\rangle$ and $|\downarrow\rangle$ are the eigenstates of the operator σ_z with eigenspins 1 and -1, respectively.
- We may also say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are the eigenvectors of the matrix σ_z with eigenvalues 1 and -1, respectively.
- In general, for an operator A which does not need to correspond to any observables/measurements, if it operates on a vector $|a\rangle$ and results in $a|a\rangle$, then we say $|a\rangle$ and a are the eigenvector and eigenvalue of A, respectively.

$$A|a\rangle = a|a\rangle$$

Eigenvalues and Eigenvectors

- What if we operate σ_z on $|\leftarrow\rangle$ and $|\rightarrow\rangle$ (i.e. $|+\rangle$ and $|-\rangle$)? Since we are using matrix representation, we need to make sure which basis we want to use so that we know what components we have in the column vector.
- Let us still use the $|\uparrow\rangle$ and $|\downarrow\rangle$ (i.e. $|0\rangle$ and $|1\rangle$) as the basis. Then we should represent $|\leftarrow\rangle$ and $|\rightarrow\rangle$ as the linear combinations of $|0\rangle$ and $|1\rangle$. We see that:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- The following shows that $|+\rangle$ and $|-\rangle$ are not the eigenvector of σ_z .

$$\sigma_z|+\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \times 1 + 0 \times 1 \\ 0 \times 1 + -1 \times 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

$$\sigma_z|-\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \times 1 + 0 \times -1 \\ 0 \times 1 + -1 \times -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$$

Eigenvalues and Eigenvectors

- We know $|+\rangle$ and $|-\rangle$ are the eigenvectors of the operator corresponding to the measurement of electron spin in the $\hat{x}$ direction. It's called σ_x . What is the matrix form of it then?
- This can be found by the following:

$$\sigma_x|+\rangle = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a \times 1 + b \times 1 \\ c \times 1 + d \times 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a + b \\ c + d \end{pmatrix} = \frac{\alpha}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \alpha|+\rangle$$

$$\sigma_x|-\rangle = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a \times 1 + b \times -1 \\ c \times 1 + d \times -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a - b \\ c - d \end{pmatrix} = \frac{\beta}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \beta|-\rangle$$

- Here, σ_x is assumed to be $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and we need to find a, b, c , and d , which are complex numbers.
- The first equation assumes $|+\rangle$ is the eigenvector of σ_x with eigenvalue α and the second equation assumes $|-\rangle$ is the eigenvector of σ_x with eigenvalue β .
- We get four equations, $a + b = \alpha$ and $c + d = \alpha$ and $a - b = \beta$ and $c - d = -\beta$.
- Physics tells us that the two eigenvalues must be 1 and -1 . This is because, in the empty space, there is no difference between up/down and left/right (we call this *isotropic*). If measuring in the $\hat{z}$ direction we get either 1 or -1 , we should expect to get 1 or -1 also if we measure in the $\hat{x}$ direction. So, we can set $\alpha = 1$ and $\beta = -1$.
- We can check that $a = d = 0$ and $b = c = 1$ are valid solutions.

Eigenvalues and Eigenvectors Finding and Phase Factor

- Now assume σ_x is given as in $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, how do we find its eigenvalues and eigenvectors?
- Assume the eigenvector to be $\begin{pmatrix} e \\ f \end{pmatrix}$ and the eigenvalue to be γ . We have

$$\begin{aligned}\sigma_x |eigenvector\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} 0 \times e + 1 \times f \\ 1 \times e + 0 \times f \end{pmatrix} = \begin{pmatrix} f \\ e \end{pmatrix} \\ &= |eigenvector\rangle = \gamma \begin{pmatrix} e \\ f \end{pmatrix}\end{aligned}$$

- Therefore,

$$f = \gamma e \quad \text{--- (eq. 1)} \quad \text{and} \quad e = \gamma f \quad \text{--- (eq. 2)}$$

- Substituting the second equation into the first one, we obtain $\gamma^2 = 1$. Therefore, the eigenvalues, γ , are ± 1 .
- Remember only normalized vectors are meaningful because the sum of the probabilities of collapsing into each state needs to be 1. So $|e|^2 + |f|^2 = 1$.
- When $\gamma = 1$, $e = f$, and therefore $|e|^2 + |f|^2 = |e|^2 + |e|^2 = 1$, and $e = f = \frac{1}{\sqrt{2}}$ is a solution.
- When $\gamma = -1$, $e = -f$, and therefore $|e|^2 + |f|^2 = |e|^2 + |-e|^2 = 1$, and $e = -f = \frac{1}{\sqrt{2}}$ is a solution.
- So, the corresponding eigenvectors are $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (i.e. $|+\rangle$) and $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (i.e. $|-\rangle$), respectively.

Eigenvalues and Eigenvectors Finding and Phase Factor

- Since e and f may be complex numbers, there are actually infinite numbers of solutions (eigenvectors). However, every solution is just different from $|+\rangle$ and $|-\rangle$ by a **phase factor**.
- Phase factor is very important in quantum computing which is used to achieve interference.
- A phase factor is a complex number with a magnitude equal to 1. Therefore, it can be written as $e^{i\theta}$, where θ is the phase. It can be expanded as:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

- However, a phase factor applying to a standalone state has no physical significance. This is because all physical observables are related to the inner products of vectors and the phase factor will disappear.
- What if $|v\rangle$ is scaled by a phase factor $e^{i\theta}$? It becomes $e^{i\theta}|v\rangle$.
- The bra version of the new vector becomes $(e^{i\theta})^* \langle v| = e^{-i\theta} \langle v|$.
- The inner product is $e^{-i\theta} \langle v| e^{i\theta} |v\rangle = \langle v|v\rangle$ because $e^{i\theta} e^{-i\theta} = 1$. So, adding a phase factor does not affect the inner product.

Remarks on Eigenvectors, Basis, and Measurement

- If $|v\rangle$ is an eigenvector of A , then $c|v\rangle$, which means $|v\rangle$ is scaled by any complex number c , is also an eigenvector of A . This is trivial because:

$$\begin{aligned} A|v\rangle &= \alpha|v\rangle \\ cA|v\rangle &= c\alpha|v\rangle \\ A(c|v\rangle) &= \alpha(c|v\rangle) \end{aligned}$$

- For matrix A , it is possible to have two different eigenvectors, $|v\rangle$ and $|u\rangle$, which are not a scalar scaling of the others ($|v\rangle \neq c|u\rangle$), having the same eigenvalue, α .

$$\begin{aligned} A|v\rangle &= \alpha|v\rangle \\ A|u\rangle &= \alpha|u\rangle \end{aligned}$$

- In such case, the eigenvectors (i.e. quantum eigenstates) are **degenerated**. Such situation is called **degeneration**. This means that the quantum system has multiple independent states that all yield the same measurement outcome (the same eigenvalue). And if this happens, any linear combination of these two vectors, $|\psi\rangle = c|v\rangle + d|u\rangle$, is also an eigenvector of the matrix and we say the two eigenvectors span an eigenspace of the system.
- The matrix corresponding to the energy observable has a very important role in the matrix form of the Schrödinger equation. Usually, it is the so-called the Hamiltonian that determines how a state (vector) evolves (transforms or rotates) with time. For example, σ_x and σ_z play an important role in rotating the spin of an electron.

Summary

- Operators as matrices: In quantum mechanics, operators are represented by matrices that determine how a quantum state evolves (time evolution/transformations in Hilbert space).
- Rotation vs scaling: Matrices generally “rotate” states in Hilbert space, but their eigenvectors are special—they are not rotated, only scaled.
- Eigenpairs: If $A|a\rangle = a|a\rangle$, then $|a\rangle$ is an eigenvector of A with eigenvalue a .
- Observable bases: For convenience, quantum computing typically uses the eigenvectors of observables as the computational basis.
- Classical readout: The eigenvalues of observables correspond to the classically measurable outcomes.
- Why it matters: Knowing what eigenvectors/eigenvalues are—and how to find them—is essential for analyzing measurements and state evolution.
- Action point: Practice computing eigenvalues/eigenvectors to build fluency for quantum problems.

Chapter 6:

Pauli Spin Matrices, Adjoint Matrix, and Hermitian Matrix

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Pauli Spin Matrices

- Any quantum mechanical state is a vector. We can decompose the state (i.e. the vector) into a linear combination of some basis vectors that span the space of interest.
- The eigenvectors of the matrices corresponding to observables are commonly used as the basis states.
- This is because after all, we need to retrieve information from the quantum computer through measurement and the states must collapse to one of the basis states, which is one of the eigenstates of the matrices corresponding to the observables.
- In a quantum computer, we commonly use observables with 2 eigenstates, which means it spans a 2D space because it is easier to implement quantum computers using two-level systems. That is the reason we still use “bit” in the qubit, as bit stands for “binary digit” like in classical computing. It is possible to use 3-level systems for quantum computing too, but this is less practical. One of the common two-level systems is the electron spin.
- We say that to perform a spin measurement, we need to apply an external magnetic field. The magnetic field can be applied in the $\hat{x}$, $\hat{y}$, $\hat{z}$ directions. Each of the setups and measurements corresponds to a matrix in matrix quantum mechanics. They are called **Pauli Spin Matrices**,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Pauli Spin Matrices

- Pauli matrices have some important properties.

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

- $\mathbf{I}$ is called the identity matrix. It is just like the “1” in scalar multiplications. Any scalar multiplied by “1” is unchanged and any matrix ($\mathbf{M}$) or vector ($|v\rangle$) multiplied by $\mathbf{I}$ is unchanged. For example,

$$\mathbf{I}|v\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} = |v\rangle$$

$$\mathbf{I}\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c & d \\ e & f \end{pmatrix} = \begin{pmatrix} c & d \\ e & f \end{pmatrix} = \mathbf{M}$$

Commutation and Anti-commutation

- There are other two important properties with Pauli matrices. The first one is about their commutation property.

$$\sigma_i \sigma_j \neq \sigma_j \sigma_i \text{ if } i \neq j$$

where i and j can be x , y , or z . We can also write it as:

$$[\sigma_i, \sigma_j] = \sigma_i \sigma_j - \sigma_j \sigma_i \neq 0 \text{ if } i \neq j$$

where $[\sigma_i, \sigma_j]$ is called the **commutator bracket** or just **commutator**.

- Pauli matrices do not commute with each other.
- Matrix is an operator that rotates a vector in the hyperspace. We apply the matrix to a vector from the left. For example, $\sigma_x \sigma_z |\nu\rangle$ means we first apply σ_z to rotate $|\nu\rangle$ and then σ_x to further rotate the rotated vector.
- Non-commuting operators also mean their measurements are incompatible. This is the **Heisenberg Uncertainty Principle** due to which we cannot measure x and z direction spins precisely at the same time.

Commutation and Anti-commutation

- The second property is their anti-commutation property,

$$\{\sigma_i, \sigma_j\} = \sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \mathbf{I}$$

where $\{\sigma_i, \sigma_j\}$ is the **anti-commutator**. It is different from commutator by just changing “−” to “+.”

- δ_{ij} is the Kronecker delta, which is one if $i = j$ and zero if $i \neq j$. It is just a scalar (and a real number either 1 or 0). Because anti-commutator is a matrix and $\mathbf{I}$ is the identity matrix, so δ_{ij} must be a scalar like the number “1.”
- For example,

$$\begin{aligned} \{\sigma_x, \sigma_z\} &= \sigma_x \sigma_z + \sigma_z \sigma_x \\ &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 2\delta_{xz} \mathbf{I} \end{aligned}$$

- Here, we introduced the **zero matrix, 0**. It is also called the null matrix. The result is correct because $\delta_{xz} = 0$ as $x \neq z$. We say that the three Pauli matrices, $\sigma_x, \sigma_y, \sigma_z$ anti-commute with one another because their anti-commutators are zero.

Spin Operator in Arbitrary Direction

- The spin operator in arbitrary directions corresponds to setting up the magnetic field in an arbitrary direction in the 3D space we are living in.
- This spin operator, when expressed as a matrix in the matrix quantum mechanics, has the eigenvalues corresponding to the possible measurement outcomes when we measure the spin in that particular direction.
- This is just like σ_z is the operator matrix with eigenvalues corresponding to the possible outcomes when performing spin measurement in the $\hat{z}$ direction.
- The spin operator, $\sigma_{\vec{n}}$, in direction $\vec{n}$ is given by $\sigma_{\vec{n}} = \vec{n} \cdot \vec{\sigma}$. $\vec{n}$ is a unit vector in a certain direction in the real 3D space that we are living in. We may represent it as:

$$\vec{n} = n_x \hat{x} + n_y \hat{y} + n_z \hat{z}$$

where $\hat{x}, \hat{y}, \hat{z}$ are the unit vector in the x, y , and z directions in the real space. It can also be written as $|\vec{n}\rangle$ or $\begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix}$ as long as we know that the basis vectors are $\hat{x}, \hat{y}, \hat{z}$ or $(|\hat{x}\rangle, |\hat{y}\rangle, |\hat{z}\rangle)$.

Spin Operator in Arbitrary Direction

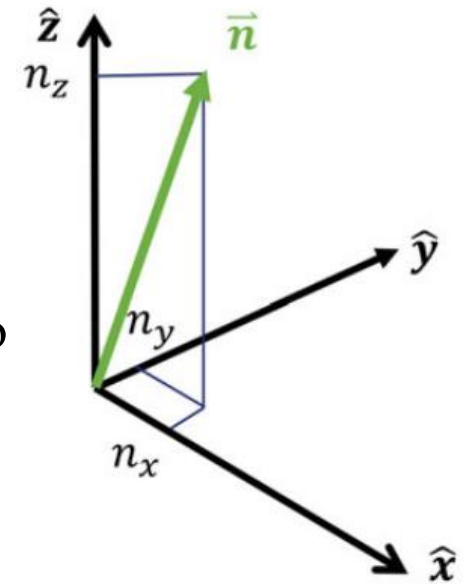
- Pic.6 tell us that $\vec{n}$ is a linear combination of $\hat{x}, \hat{y}, \hat{z}$ with coefficients n_x, n_y, n_z , respectively.
- Since $\vec{n}$ is a unit vector, which means it is normalized, $\langle \vec{n} | \vec{n} \rangle = 1$, and based on the inner product definition, it means $(n_x \ n_y \ n_z) \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix} = n_x^2 + n_y^2 + n_z^2 = 1$.
- Then how about $\vec{\sigma}$? Is it a matrix like σ_z ? But it has an arrow on the top, is it a vector?
- $\vec{\sigma}$ is a vector but the coefficients are matrix. The definition of $\vec{\sigma}$ is as following:

$$\begin{aligned} \vec{\sigma} &= \sigma_x \hat{x} + \sigma_y \hat{y} + \sigma_z \hat{z} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \hat{x} + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \hat{y} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \hat{z} \end{aligned}$$

- It is called the **Pauli vector**. Since $\vec{\sigma}$ is a vector, we can write it in column form also

as long as we know the basis vectors are $\hat{x}, \hat{y}, \hat{z}$,

$$\vec{\sigma} = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{pmatrix} = |\vec{\sigma}\rangle.$$



Pic.6 Decomposition of a unit vector $\vec{n}$ in the 3D real space.

Spin Operator in Arbitrary Direction

- Therefore, the spin operator $\sigma_{\vec{n}}$ is just an inner product between the vector $\vec{n}$ and the vector $\vec{\sigma}$.

$$\begin{aligned}
 \sigma_{\vec{n}} &= \vec{n} \cdot \vec{\sigma} = \langle \vec{n} | \vec{\sigma} \rangle \\
 &= (n_x \ n_y \ n_z) \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{pmatrix} \\
 &= n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & n_x \\ n_x & 0 \end{pmatrix} + \begin{pmatrix} 0 & -in_y \\ in_y & 0 \end{pmatrix} + \begin{pmatrix} n_z & 0 \\ 0 & -n_z \end{pmatrix} \\
 &= \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}
 \end{aligned}$$

- What if $\vec{n}$ is pointing at the $\hat{y}$ direction? It means $n_x = 0$, $n_y = 1$, and $n_z = 0$. We get $\sigma_{\vec{n}} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y$.

Relationship Between Spin Direction and Real 3D Space

- The basis states of the spin spaces (e.g. $|\uparrow\rangle$, $|\downarrow\rangle$, $|\leftarrow\rangle$, $|\rightarrow\rangle$) are not the same as the real 3D space basis ($\hat{x}$, $\hat{y}$, $\hat{z}$). They belong to different spaces and they do not talk to one another.
- In an isotropic space, electrons with different spin states, $|\psi\rangle$, have the same energy.
- When a magnetic field is applied in a certain direction (e.g. $\hat{z}$), two particular states become the eigenstates (i.e. basis vectors) of the new system with that magnetic field. We call these two states $|\uparrow\rangle$ and $|\downarrow\rangle$.
- These two states have different energies under this external magnetic field in the $\hat{z}$ direction. We can just call these two states $|0\rangle$ and $|1\rangle$.
- Moreover, it also corresponds to the classical spin direction which will give the corresponding energy change.
- Therefore, we may say that the basis states such as $|\uparrow\rangle$, $|\downarrow\rangle$, $|\leftarrow\rangle$, $|\rightarrow\rangle$ do have a relationship to the direction in the real space. However, any linear combination of them loses the connection to the real 3D space. For example, $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ does *not* correspond to $\hat{z} + \widehat{-z}$.

Adjoint and Hermitian Matrices

- We say every vector in the ket space has a corresponding vector in the bra space.
- An operator acts on a vector by rotating it in the hyperspace and the result is still a vector (after all, this is just a matrix-vector multiplication, which should result in a vector).
- So, what will happen to its corresponding vector in the bra space?
- Mathematicians tell us that the corresponding operator in the bra space is its **adjoint matrix**. If the matrix is $\mathbf{A}$, we denote its adjoint matrix as $\mathbf{A}^\dagger$.
- The way to find the adjoint matrix of $\mathbf{A}$ is the same as finding the corresponding vector in the bra space for a given vector in the ket space, i.e. by applying conjugation, $\mathbf{A}^*$, and transpose, $(\mathbf{A}^*)^T$, operations.

$$\mathbf{A}^\dagger = \mathbf{A}^{*T}$$

$$A_{ji}^\dagger = A_{ij}^*$$

- If $\mathbf{A} = \begin{pmatrix} i & i \\ 0 & 1 \end{pmatrix}$, then $\mathbf{A}^* = \begin{pmatrix} -i & -i \\ 0 & 1 \end{pmatrix}$ and $(\mathbf{A}^*)^T = \begin{pmatrix} -i & 0 \\ -i & 1 \end{pmatrix}$. Therefore, $\mathbf{A}^\dagger = \begin{pmatrix} -i & 0 \\ -i & 1 \end{pmatrix}$.
- By choosing $i = 0, j = 1$, $A_{01}^* = -i$ and $A_{10}^\dagger = -i$ as expected.

Adjoint and Hermitian Matrices

- How do we operate the matrix on a bra vector? While an operator operates on ket vector from the *left*, it operates from the *right* on the bra vector.
- If $|v\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix}$, $A|v\rangle = \begin{pmatrix} i & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -1+i \\ 1 \end{pmatrix}$. The corresponding vector of $A|v\rangle$ in the bra space is thus $(-1 - i \ 1)$, after taking the transpose and conjugation operations.
- We can also apply $A^\dagger$ to $\langle v|$ directly from the *right*. Since $\langle v| = (-i \ 1)$, $\langle v|A^\dagger = (-i \ 1) \begin{pmatrix} -i & 0 \\ -i & 1 \end{pmatrix} = (-1 - i \ 1)$. The results are the same.
- What if we take the adjoint of the adjoint of a matrix? It reverts to itself, $(A^\dagger)^\dagger = A$.
- If a matrix's adjoint matrix equals itself, this matrix is **self-adjoint** or **Hermitian**, $A^\dagger = A$ and $A_{ji}^\dagger = A_{ji}$.
- Then $A_{ij}^* = A_{ji}$, it means that after performing the transpose and complex conjugation, the matrix is still the same. This is similar to a real number, in which after performing complex conjugation, it still equals itself.

Summary

- Always distinguish scalars, vectors, and matrices in the notation.
- Coefficients can themselves be matrices (e.g., in the Pauli vector).
- Pauli matrices are fundamental in quantum mechanics and quantum computing—try to memorize them.
- We also discussed the adjoint ($\dagger$) and Hermitian matrices.